

Fraud Application Parameters — Explained

Reference: Bank Account Fraud (BAF) style feature set — Application ID: APP-LOW-plmoknij

This record resembles an entry from a bank-account-opening fraud detection dataset (similar to the well-known 'Bank Account Fraud' / IEEE-CIS style feature schemas). Below is an explanation of each field, grouped by category, followed by a brief assessment of this specific record.

Applicant / Identity Signals

Parameter	Value	Explanation
income	0.7	Applicant's annual income, given as a normalized/percentile value (0–1) rather than a raw dollar figure.
name_email_similarity	0.85	A score measuring how closely the applicant's name matches their email address (high = name and email look consistent, e.g. "john.smith@..." for John Smith).
customer_age	35	Applicant's age (often bucketed in the original dataset, e.g. rounded to nearest 10).
date_of_birth_distinct_emails_4w	1	Number of distinct email addresses associated with this same date of birth in the past 4 weeks — a way to catch identity reuse/fraud rings.

Address History

Parameter	Value	Explanation
prev_address_months_count	12	How many months the applicant lived at their previous address.
current_address_months_count	60	How many months at their current address (longer = more stable/trustworthy signal).

Application Timing / Behavior

Parameter	Value	Explanation
days_since_request	0.007	Time elapsed since the application was submitted (very small = essentially real-time).
intended_balance_amount	50	The balance/amount the applicant intends to transfer into the new account.
session_length_in_minutes	10.5	How long the applicant spent in the application session.
keep_alive_session	1	Whether the session was kept active/alive rather than idling out (1 = yes).

Velocity Features (fraud-ring indicators)

Parameter	Value	Explanation
zip_count_4w	900	Number of applications from the same ZIP code in the last 4 weeks.
velocity_6h	1300	Application velocity (rate of similar applications) over the past 6 hours.
velocity_24h	1600	Same, over 24 hours.
velocity_4w	1900	Same, over 4 weeks.

Banking Relationship

Parameter	Value	Explanation
bank_branch_count	3	Number of bank branches associated with the applicant (or applications tied to that branch, depending on dataset version).
bank_months_count	36	How long (months) the applicant has had a relationship with the bank.
has_other_cards	1	Whether the applicant already holds other credit cards with this bank (1 = yes).
credit_risk_score	190	An internal/external credit risk score.
proposed_credit_limit	400	The credit limit being proposed for this account.

Payment / Employment / Housing (Categoricals)

Parameter	Value	Explanation
payment_type	AB	Anonymized category for the payment plan type.
employment_status	CB	Anonymized employment status category (e.g., employed, self-employed, unemployed — codes are anonymized).
housing_status	BC	Anonymized housing status category (e.g., own, rent, other).

Contact Verification

Parameter	Value	Explanation
email_is_free	0	Whether the email is from a free provider like Gmail/Yahoo (0 = no, it's a paid/custom domain).
phone_home_valid	1	Whether the home phone number passed validation (1 = valid).
phone_mobile_valid	1	Whether the mobile phone number passed validation (1 = valid).

Device / Fraud Signals

Parameter	Value	Explanation
foreign_request	0	Whether the request originated from a foreign country (0 = no).
source	INTERNET	Channel through which the application was submitted (internet vs. app, etc.).
device_os	macintosh	Operating system of the device used.
device_distinct_emails_8w	1	Number of distinct emails seen on this device in the past 8 weeks (high = possible device shared across fake identities).
device_fraud_count	0	Count of prior fraud flags associated with this device.

Metadata

Parameter	Value	Explanation
month	7	Month the application was made (July).
application_id	APP-LOW-p lmoknij	A unique identifier for this application.

Assessment of This Record

Most signals in this record look clean — valid phones, long address/bank tenure, and a sensible name-email match. However, the **velocity features are quite high** (1,300–1,900) and **zip_count_4w is 900**, which are the kind of values a fraud model would weight heavily. These suggest unusually high application activity clustered around this profile's ZIP code, device, or identity pattern — often the strongest signal of fraud in this type of dataset.